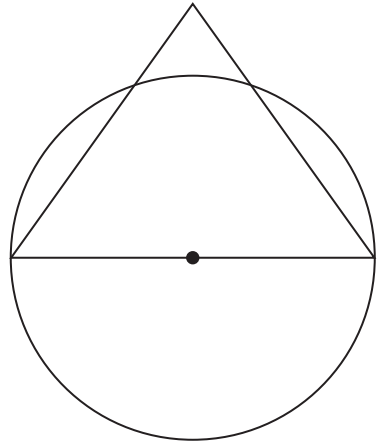
13 The diagram shows a circle and an equilateral triangle.



One side of the equilateral triangle is a diameter of the circle.
The circle has a circumference of 44 cm.

Work out the area of the triangle.
Give your answer correct to 3 significant figures.

Circumference = πd

$44 = \pi d$, so $d = 44 \div \pi$

$d = 14.00563499...$ cm

1 mark
Find the side length from the circumference

Equilateral triangle: every side is d, every angle is 60°.

$$\text{Area} = \frac{1}{2} \times d \times d \times \sin 60°$$

1 mark
Use ½ab sin C

$= ½ \times (44 \div \pi)^2 \times \sin 60°$

$= 84.9388239...$ cm^2

1 mark
Round only the final answer to 3 significant figures

84.9 cm^2

(Total for Question 13 is 3 marks)